

JaxMARL-HFT: GPU-Accelerated Large-Scale Multi-Agent Reinforcement Learning for High-Frequency Trading

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J.P.Morgan

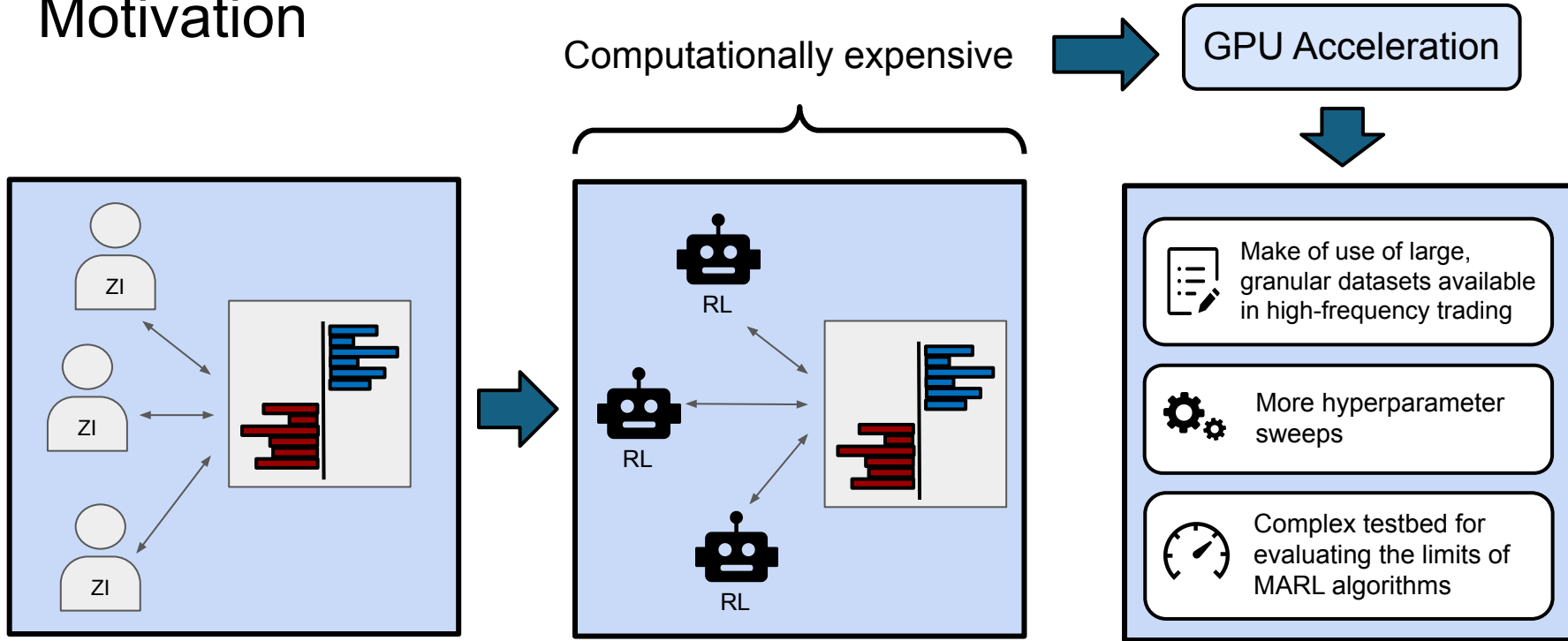
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Motivation

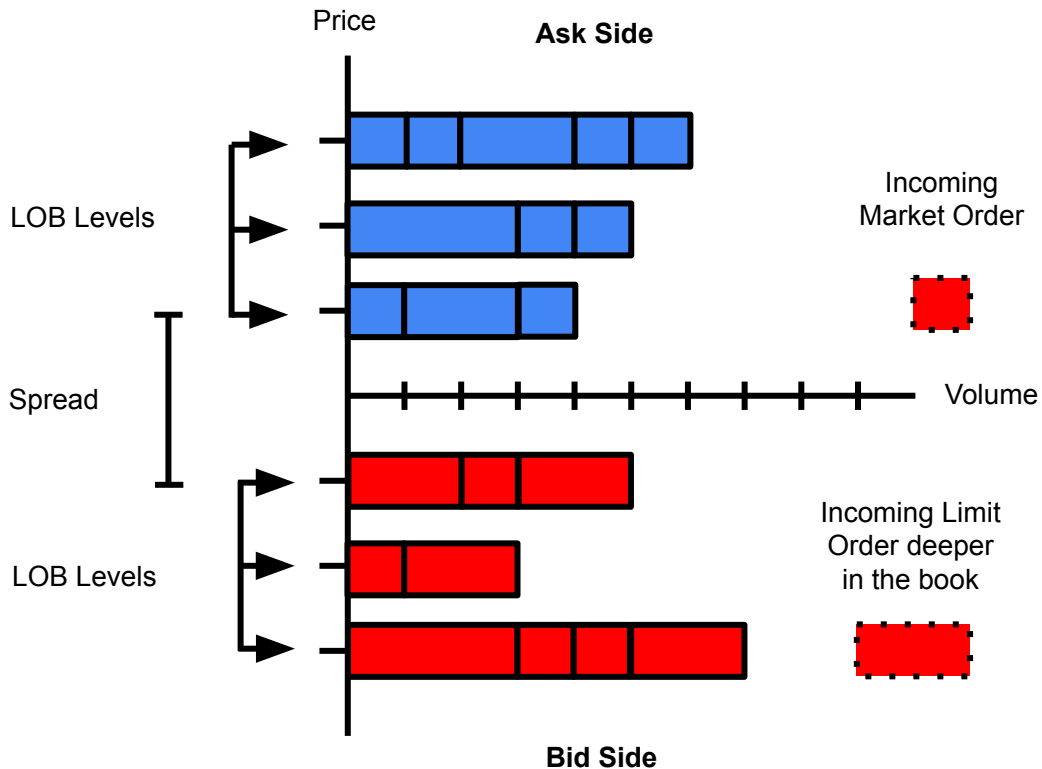


Moving agent-based modelling approaches for financial markets from heuristic (zero-intelligence) agents to intelligent reinforcement learning agents

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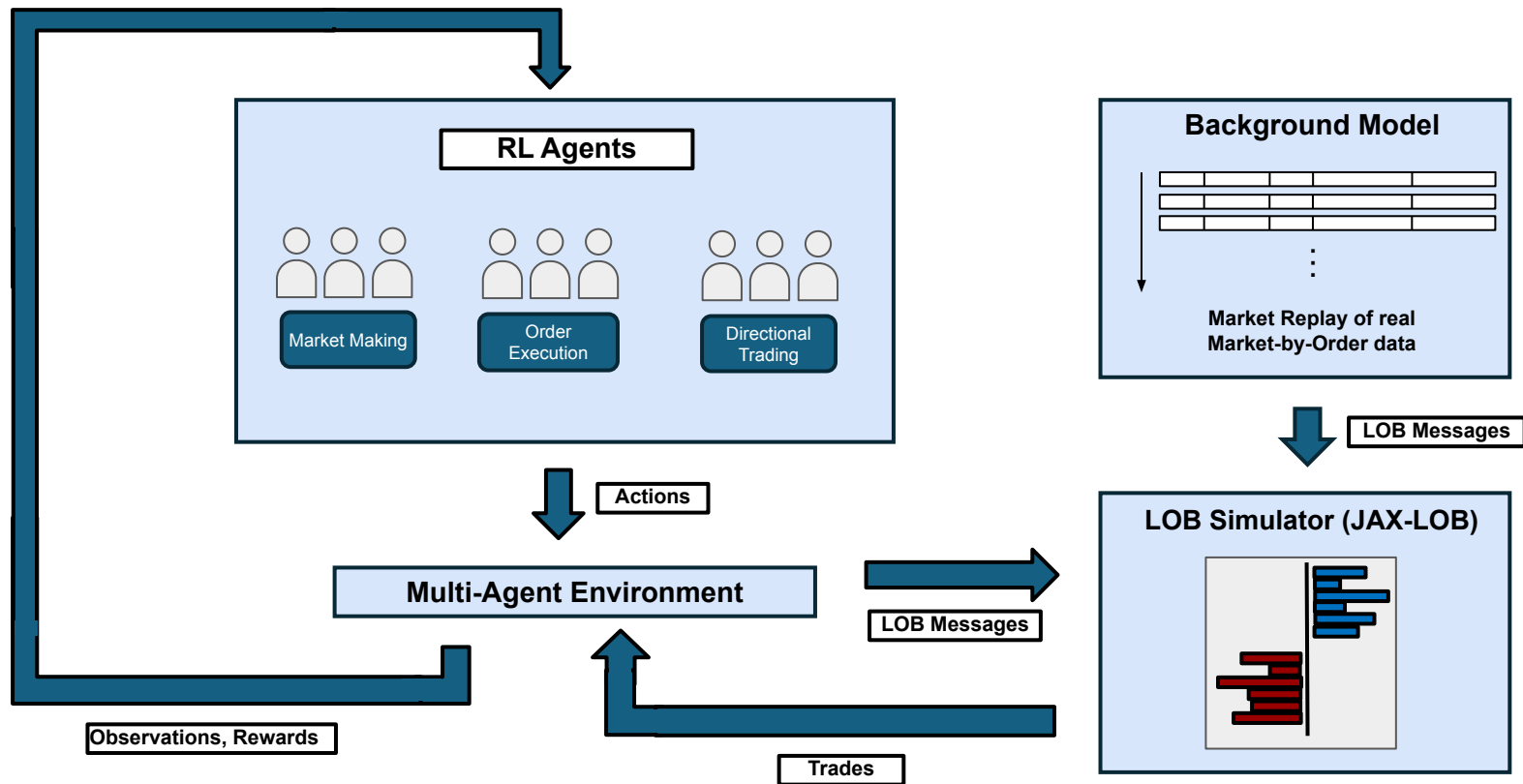
Background: Limit Order Books



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The JaxMARL-HFT Framework

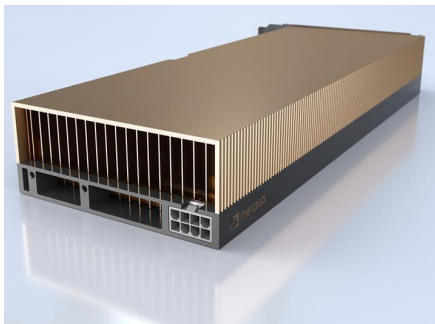


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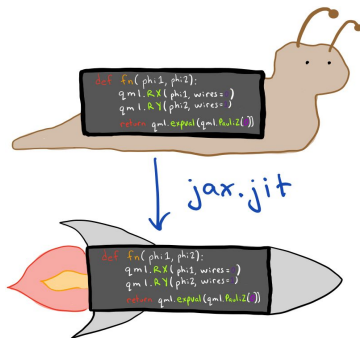
GPU-Acceleration

JAX code runs natively on GPU which allows for massive parallelization



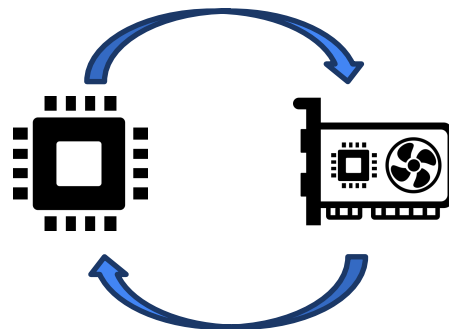
<https://www.nvidia.com/en-us/data-center/a40/>

Just in Time (JIT) compilation with XLA (Accelerated Linear Algebra)

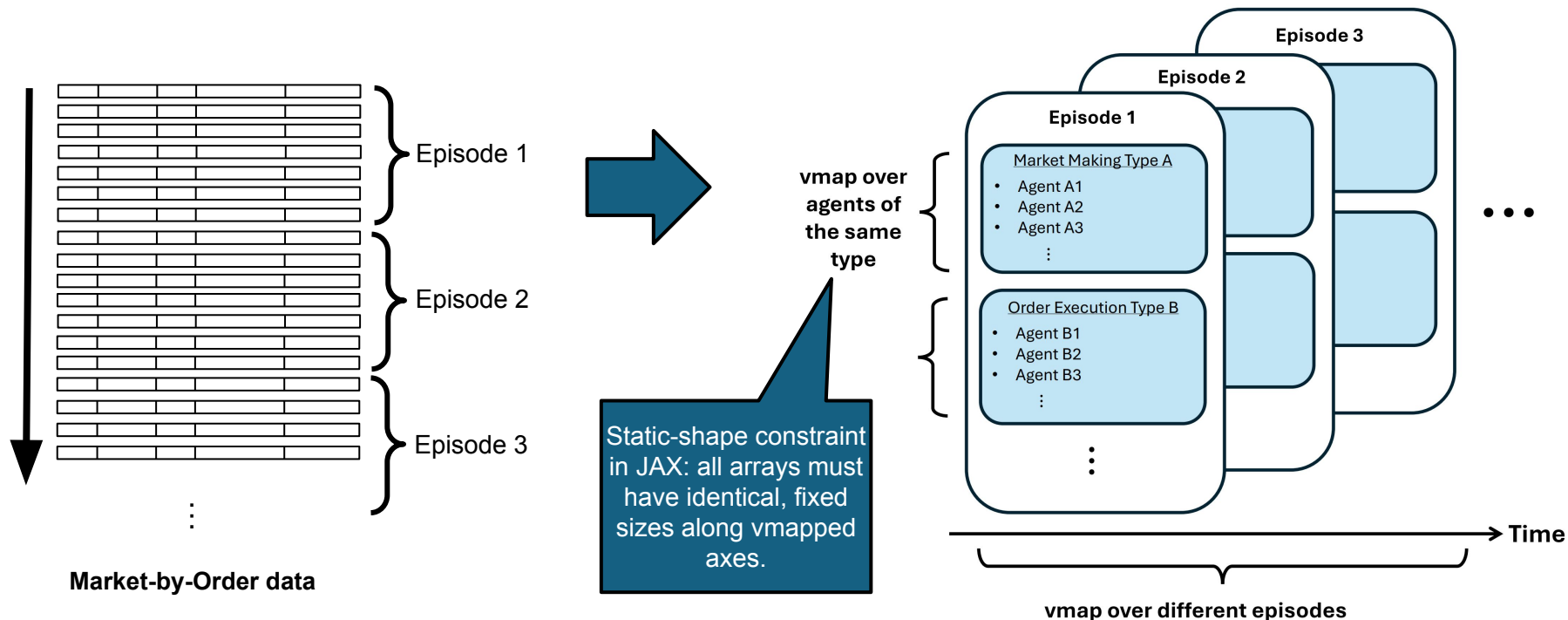


https://pennylane.ai/qml/demos/tutorial_jax_transformations/

Computations remain on the GPU, avoiding costly CPU–GPU transfers



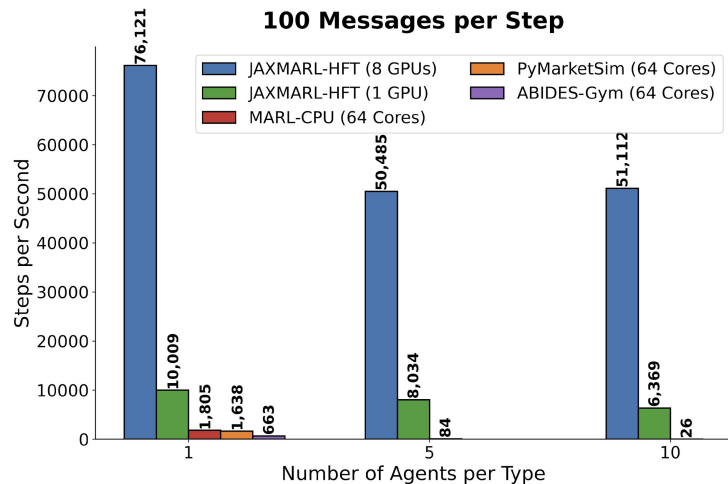
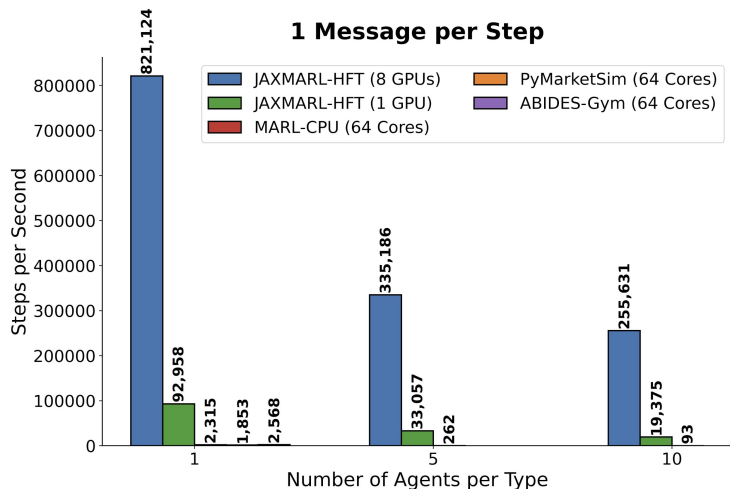
GPU-Acceleration in JaxMARL-HFT: Two Levels of Parallelization



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Reinforcement Learning Training Speed



Benefit: up to 240x speed-up over state-of-the-art reference implementations

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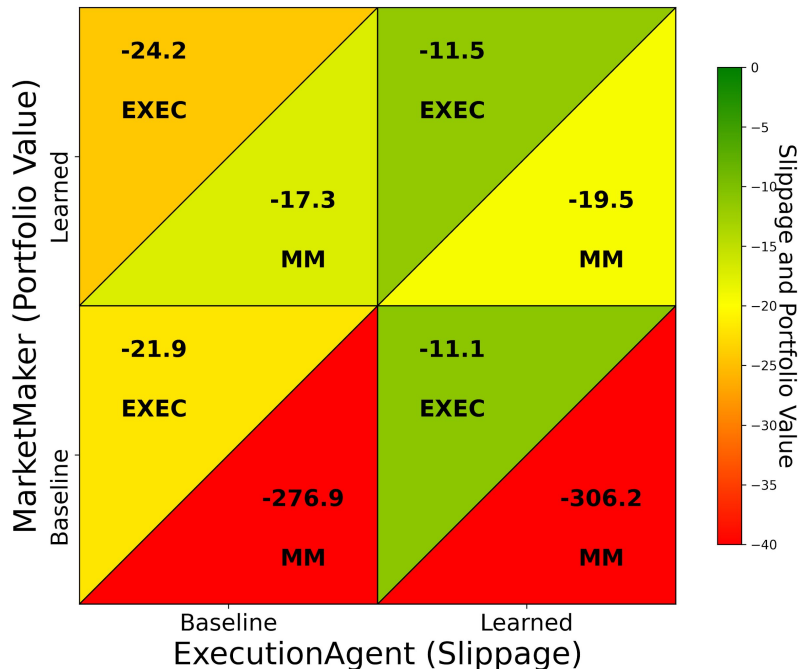
Independent PPO Training Result

Two player: market making and order execution

Both agents outperform their respective baselines

Each agent performs worse when interacting with a learned counterpart compared to when facing a baseline policy

Slippage and Portfolio Value: Baseline vs Learned



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Conclusion & Future Work

Summary



GPU acceleration enables large-scale MARL experiments through substantial speed-ups



Real-world relevant testbed for MARL as it is fully compatible with state-of-the-art algorithms implemented in JaxMARL [1]



Future Work



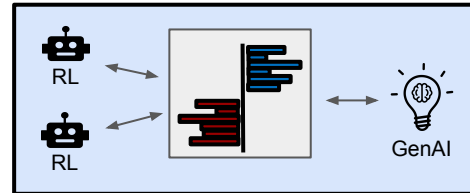
Opponent shaping



Market Impact Analysis



General-purpose action/observation spaces



Replacing the static MBO background model with a generative model such as in Nagy et al. [2]



Evaluate realism of data distributions (MARL, GenAI, hybrid) using LOB-Bench [3]

[1] Alexander Rutherford et al. 2024. JaxMARL: Multi-agent RL Environments and Algorithms in JAX. *Advances in Neural Information Processing Systems* 37 (2024), 50925–50951

[2] Peer Nagy, Sascha Frey, Silvia Sapora, Kang Li, Anisoara Calinescu, Stefan Zohren, and Jakob Foerster. 2023. Generative AI for end-to-end limit order book modelling: A token-level autoregressive generative model of message flow using a deep state space network. In *Proceedings of the Fourth ACM International Conference on AI in Finance (ICAIF '23)*. 91–99

[3] Peer Nagy, Sascha Frey, Kang Li, Bidipta Sarkar, Svitlana Vyetenko, Stefan Zohren, Anisoara Calinescu, and Jakob Foerster. 2025. *LOB-Bench: Benchmarking Generative AI for Finance — an Application to Limit Order Book Data*. In *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*

Questions



GitHub Repository



Arxiv Paper



Additional Slides

Different Agent Types

Market Making

Reward based on Spooner (2018) [1]:

$$r_{Sp} = \Psi_b + \Psi_s + \overbrace{\Psi_{INV} - (1 - \lambda) \max(0, \Psi_{INV})}^{\text{Asymmetrically dampened Inventory PnL}}$$

Buy PnL Term: $\Psi_b = \sum (\bar{M} - P_b) Q_b$

Sell PnL Term: $\Psi_s = \sum (P_a - \bar{M}) Q_a$

Inventory PnL Term: $\Psi_{INV} = I_t (M_t - M_{t-1})$

Average mid price of the previous step: $\bar{M}$

Buy prices of the agent: P_b

Quantity bought: Q_b

Inventory: I_t

Mid price change during the step: $M_t - M_{t-1}$

Order Execution

$$r = \sum_j Q_j (P_j - P_{VWAP})$$

Volume-weighted average price of the step based on non-agent trades: P_{VWAP}

Directional Trading

$$r = PV = Q_{inv} \times P_{ref} + Cash$$

Quantity of the inventory: Q_{inv}

Reference price (mid price or best bid/ask): P_{ref}